

Text Classification Pipeline Forensics: Reproducibility, Shortcut Sensitivity, and Data Quality in Disaster Tweets

LAI Jiaxing

HE Jiarui

Abstract—This project investigates when apparent improvements to a disaster-tweet classifier are reproducible and trustworthy rather than products of split drift, superficial shortcuts, operating-point changes, or label inconsistency. All experiments used 7,613 labeled tweets and one immutable, stable-ID partition of 4,567 training, 1,523 development (dev), and 1,523 held-out examples. Ticket decisions were made from train/dev evidence, frozen, and only then reported on held-out data. A raw-text word-unigram TF-IDF plus Logistic Regression baseline achieved held-out target-class F1 of 0.749186, missing the supplied reference 0.757422 by 0.008236, although the local pipeline was deterministic and leakage-controlled. URL placeholder produced a small, precision-oriented improvement and exact invariance under a paired URL perturbation. A higher-scoring shallow-feature model was rejected because masking keyword metadata changed 702 dev predictions and reduced F1 from 0.749396 to 0.642306. Balanced Logistic Regression was selected on dev as a recall-oriented operating point; on held-out it achieved precision 0.744361, recall 0.756881, F1 0.750569, and accuracy 0.783979, while fixing 35 baseline false negatives and introducing 56 false positives. A subsequent duplicate and label audit found substantial cross-split relationships and conflicting annotations, but an eight-row in-memory correction probe reduced dev F1 and was rejected. The final frozen model therefore retains the balanced classifier with no label corrections. Five clean-process replays reproduced labels and metrics with maximum score drift 1.11×10^{-16} . The results show that score changes require transition, robustness, provenance, and case evidence; they do not establish universal deployment superiority.

Index Terms—text classification, reproducibility, error analysis, dataset shortcuts, decision thresholds, data quality

I. PROJECT PROBLEM AND GOAL

A. Problem Context and Forensic Goal

The task is binary classification of the Kaggle Disaster Tweets data: target 1 denotes a tweet labeled as describing a real disaster and target 0 denotes a tweet labeled otherwise [1]–[3]. The assignment is deliberately forensic. A higher aggregate score can arise because a transformation removed nuisance variation, because a decision threshold happened to fit the metric, because metadata exposed a benchmark shortcut, or because inconsistent labels changed what the metric rewards. Score differences alone therefore cannot determine whether a pipeline change is trustworthy.

The project goal was to connect each proposed change to four kinds of evidence: a controlled intended lever, dev-set metric and transition behavior, row-level examples with stable IDs, and a frozen post-selection held-out report. Five tickets

covered baseline reproduction, text normalization, shallow-feature shortcuts, decision rules and classifier choices, and data quality. The unifying question was not simply which row of a score table was largest, but which conclusions remained defensible when the prediction movements, perturbation behavior, data provenance, and unresolved uncertainty were considered together.

This report distinguishes observation from interpretation. Metrics, counts, scores, IDs, and hashes are observations regenerated from repository artifacts. Conclusions such as “shortcut-sensitive,” “recall-oriented,” or “potential annotation issue” are bounded interpretations of those observations. They are not causal claims and do not replace external label adjudication.

B. Evidence Contributions

The project contributes: (1) a deterministic majority floor and raw-text TF-IDF baseline audit; (2) six isolated normalization comparisons plus a raw control; (3) ten text, metadata, and shallow-feature comparisons with targeted perturbations; (4) a bounded threshold, regularization, class-weight, and LinearSVC study; (5) exact, canonical, and bounded near-duplicate analyses with a controlled label-correction probe; and (6) a frozen-decision replay audit. All prediction transitions use Ticket 1 as the common comparator, as required by the instructor clarification [4]. These are project-specific contributions on one fixed benchmark, not claims of a novel classifier or broad external generalization.

II. METHODOLOGY

A. Dataset and Fixed Split

The labeled CSV contains the fields `id`, `keyword`, `location`, `text`, and `target`. The 7,613 IDs are unique. The instructor-provided split assigns each stable ID exactly once to train, dev, or held-out, with no ID overlap. Table I gives the class counts. Train is the only fitting partition. Dev is the only partition used to choose preprocessing, features, hyperparameters, class weights, or thresholds. Held-out is restricted to reporting after a ticket decision is frozen [5], [6].

The split is stratified with seed 3102. ID-based joins were used throughout; dataframe position was never treated as identity. This prevents row-order changes from silently changing a comparison. The source-data SHA-256 is 61111c6d...60df, and the split SHA-256 is

TABLE I
DATASET AND FIXED SPLIT STATISTICS

split	rows	target 0	target 1	target 1 rate	unique ids	seed
train	4567	2605	1962	0.429604	4567	3102
dev	1523	868	655	0.430072	1523	3102
heldout	1523	869	654	0.429416	1523	3102

Counts are tweets; target-1 rate is a fraction in $[0,1]$. The split is fixed and stratified.

db2fd1fd...88f1; full values remain in the frozen manifest. ID disjointness does not imply semantic independence, however: Ticket 5 found exact and approximate text relationships across splits.

B. Metrics and Common Comparator

Target 1 is the positive class. For true positives TP , false positives FP , false negatives FN , and true negatives TN , the reported measures are

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad (1)$$

$$F_1 = \frac{2PR}{P + R}, \quad A = \frac{TP + TN}{TP + TN + FP + FN}. \quad (2)$$

All metric values are unitless. Confusion and transition entries are tweet counts. The default probability rule is inclusive at threshold 0.50. A *fixed FP* is a target-0 row predicted 1 by Ticket 1 and 0 by a candidate; a *new FP* moves in the opposite direction. Fixed and new FNs are defined analogously for target-1 rows. These categories describe different semantic costs and are never netted into a single “errors changed” value.

F1 is the primary declared selection metric, but it contains no deployment-specific cost model. Accuracy and the full confusion matrix are retained so that recall gains cannot be presented without their precision and false-positive costs. Stable-ID change tables supply the complementary case evidence.

C. Selection, Freeze, and Held-Out Policy

Each ticket followed the same chronology: predeclare the question and bounded candidates; fit only on train; evaluate and interpret on dev; write an immutable decision artifact; then perform or reuse a post-freeze held-out report. Every frozen record states that held-out was not used for selection and that reopening was not permitted. Tickets 3 and 5 intentionally reuse previously validated prediction cores because their dev decisions retained an earlier model. This avoids unnecessary refitting while preserving ticket provenance.

Two later actions are explicitly separated from model selection. First, after a documented rollback removed Ticket 1 row-level artifacts, one authorized recovery replay reconstructed them with the exact frozen recipe. The historical primary comparison count remained one, and no parameter was changed. Second, after all five decisions were frozen, five distinct clean processes replayed the final recipes for audit only. Neither action reopened a decision. Procedural provenance is important because artifacts from earlier tickets existed during later work; the protection is a documented freeze policy, not a claim that every prior score was literally unknown.

D. Floor, Baseline, and Reproducibility Controls

The floor predicts the train-majority class, target 0, for every row. On dev it therefore has target-1 precision, recall, and F1 of zero, accuracy 0.569928, and confusion counts 868/0/655/0 for $TN/FP/FN/TP$. Its role is wiring validation rather than competition.

The stronger baseline uses raw tweet text only, default lowercase word-unigram TF-IDF, and Logistic Regression with $C = 1$, no class weighting, the `lbfgs` solver, `max_iter=100`, threshold 0.50, and random state 3102. No metadata, manual normalization, threshold tuning, or train-plus-dev vocabulary fitting is used. Two independent baseline dev fits produced identical IDs, labels, scores, metrics, warnings, and iteration counts; the selected fit converged in 15 iterations. Table II separates the dev floor and baseline from the later reference comparison.

The locked audit environment recorded CPython 3.13.0, scikit-learn 1.9.0, pandas 3.0.3, NumPy 2.5.1, Matplotlib 3.11.1, and one CPU thread where supported [7]–[10]. Locked local reproducibility does not imply byte-identical probabilities on arbitrary systems or future package versions.

E. Controlled Ticket Design

Tickets 2–5 changed one intended lever or one explicitly declared bundle while retaining the fixed split and comparator. Ticket 2 enabled one surface transformation at a time. Ticket 3 isolated metadata families and declared text-feature combinations. Ticket 4 retained the raw text representation while changing one threshold, regularization, class-weight, or classifier lever. Ticket 5 changed only eight proposed training labels in an in-memory candidate; duplicate discovery and audit dispositions were not hidden training interventions. Candidate grids were bounded and evaluated on one fixed dev split, so they support controlled comparisons rather than exhaustive optimization.

III. MAIN EVIDENCE AND RESULTS

A. Cross-Ticket Quantitative Overview

Table III summarizes the five frozen decisions. Tables IV and V give the full selected dev and held-out metrics; Table VI makes the error-count tradeoffs explicit. Table VII reports every ticket relative to the same Ticket 1 comparator. Ticket 1 equals Ticket 3 by deliberate retention, and Tickets 4 and 5 share a prediction core because Ticket 5 rejected corrections. These repeated rows are provenance facts rather than independent experimental replications.

The tables already show why post-hoc held-out ranking would be invalid. Ticket 2 has the largest observed held-out

TABLE II
BASELINE FLOOR AND ASSIGNMENT-REFERENCE COMPARISON

condition	split	metric	value	reference or tolerance	absolute gap	within tolerance
train_majority_floor	dev	f1_target_1	0	—	—	—
raw_text_tfidf_logistic_regression	dev	f1_target_1	0.738812	—	—	—
frozen_baseline_heldout	heldout	heldout_f1_target_1	0.749186	0.757422	0.008236	False

F1 is unitless in [0,1]. The assignment reference tolerance is 0.001; the frozen held-out result does not match it.

TABLE III
PROJECT-WIDE RESULTS ACROSS FIVE TICKETS

ticket	model name	dev f1	heldout f1	heldout accuracy	fixed fp	fixed fn	new fp	new fn
1	raw_text_tfidf_logistic_regression	0.738812	0.749186	0.797768	0	0	0	0
2	normalize_urls_placeholder	0.740313	0.753117	0.804990	22	8	4	15
3	raw_text_tfidf_logistic_regression	0.738812	0.749186	0.797768	0	0	0	0
4	lr_c1_balanced_default	0.752085	0.750569	0.783979	0	35	56	0
5	lr_c1_balanced_default	0.752085	0.750569	0.783979	0	35	56	0

Metrics are recomputed from archived predictions. Transition counts are held-out changes relative to Ticket 1; counts are tweets.

TABLE IV
SELECTED DEV METRICS BY TICKET

ticket	precision target 1	recall target 1	f1 target 1	accuracy	true negative	false positive	false negative	true positive
1	0.790941	0.693130	0.738812	0.789232	748	120	201	454
2	0.804659	0.685496	0.740313	0.793171	759	109	206	449
3	0.790941	0.693130	0.738812	0.789232	748	120	201	454
4	0.746988	0.757252	0.752085	0.785292	700	168	159	496
5	0.746988	0.757252	0.752085	0.785292	700	168	159	496

Precision, recall, F1, and accuracy are unitless in [0,1]; confusion entries are tweet counts. Selection used dev evidence.

TABLE V
FROZEN HELD-OUT METRICS BY TICKET

ticket	precision target 1	recall target 1	f1 target 1	accuracy	true negative	false positive	false negative	true positive
1	0.801394	0.703364	0.749186	0.797768	755	114	194	460
2	0.825137	0.692661	0.753117	0.804990	773	96	201	453
3	0.801394	0.703364	0.749186	0.797768	755	114	194	460
4	0.744361	0.756881	0.750569	0.783979	699	170	159	495
5	0.744361	0.756881	0.750569	0.783979	699	170	159	495

Metrics are descriptive held-out results after dev-only freezing; they were not used to reopen selection.

F1, 0.753117, whereas the final Ticket 5 core has 0.750569. Ticket 2 cannot be selected after that observation because its normalization decision and the later tickets answer different predeclared questions. The procedurally valid final choice is the last dev-frozen decision: Ticket 5 retaining Ticket 4 with no corrections.

B. Ticket 1: Baseline Discrepancy Diagnosis

Hypothesis and lever. The pre-freeze hypothesis was that a literal, version-locked raw-text default pipeline would reproduce the supplied reference within tolerance. Ticket 1 first froze that minimal pipeline. Diagnostic probes then changed casing, word n-grams, solver, regularization C , iteration cap, class weighting, seed, explicit L2 spelling, or TF-IDF fit scope one at a time. Dev evidence was the diagnostic basis, and no probe was allowed to rewrite the frozen baseline after the reference result was observed. A preserved historical ledger also contains post-mismatch held-out metrics for the baseline and 32 probes; this descriptive record is reported separately and has no selection authority.

Dev evidence and decision. The selected dev baseline achieved precision 0.790941, recall 0.693130, F1 0.738812, and accuracy 0.789232 (Table IV). Lowercasing off reduced F1 to 0.731148; word bigrams to 0.728745; liblinear to

0.737785; $C = 0.5$ to 0.726672; and an intentionally invalid train-plus-dev TF-IDF fit to 0.725237. Increasing `max_iter`, changing the seed to 9999, and spelling L2 explicitly caused no prediction change. Balanced weights raised dev F1 to 0.752085, but this was a post-freeze diagnostic and became an eligible lever only in Ticket 4. The decision was to retain the minimal baseline.

Held-out evidence. The frozen model achieved precision 0.801394, recall 0.703364, F1 0.749186, accuracy 0.797768, and 755/114/194/460 for $TN/FP/FN/TP$. The supplied reference was 0.757422 with tolerance 0.001; the absolute gap 0.008236 exceeded that tolerance (Table II). The local implementation is deterministic, converged, stable-ID aligned, and train-only, but the external reference provides no predictions, effective parameter dictionary, preprocessing code, solver diagnostics, or package versions. Consequently, no particular solver, version, or preprocessing setting can be identified as the cause.

Historical probe landscape. Table VIII and Figure 1 join the 33 saved historical probe records: the submitted baseline plus 32 single-lever variants. Each coordinate is a baseline-relative F1 change in percentage points, with dev on the horizontal axis and held-out on the vertical axis. The left

TABLE VI
SELECTED DEV AND HELD-OUT CONFUSION-MATRIX COUNTS

ticket	split	model name	tn	fp	fn	tp
1	dev	raw_text_tfidf_logistic_regression	748	120	201	454
1	heldout	raw_text_tfidf_logistic_regression	755	114	194	460
2	dev	normalize_urls_placeholder	759	109	206	449
2	heldout	normalize_urls_placeholder	773	96	201	453
3	dev	raw_text_tfidf_logistic_regression	748	120	201	454
3	heldout	raw_text_tfidf_logistic_regression	755	114	194	460
4	dev	lr_c1_balanced_default	700	168	159	496
4	heldout	lr_c1_balanced_default	699	170	159	495
5	dev	lr_c1_balanced_default	700	168	159	496
5	heldout	lr_c1_balanced_default	699	170	159	495

TN/FP/FN/TP use target 1 as the positive class; counts are tweets.

TABLE VII
PREDICTION TRANSITIONS RELATIVE TO THE TICKET 1 FROZEN BASELINE

ticket	split	baseline	fixed fp	fixed fn	new fp	new fn
1	dev	ticket_1_frozen	0	0	0	0
1	heldout	ticket_1_frozen	0	0	0	0
2	dev	ticket_1_frozen	22	6	11	11
2	heldout	ticket_1_frozen	22	8	4	15
3	dev	ticket_1_frozen	0	0	0	0
3	heldout	ticket_1_frozen	0	0	0	0
4	dev	ticket_1_frozen	0	42	48	0
4	heldout	ticket_1_frozen	0	35	56	0
5	dev	ticket_1_frozen	0	42	48	0
5	heldout	ticket_1_frozen	0	35	56	0

Fixed/new FP/FN are disjoint correctness transitions; counts are tweets and are reported separately for dev and held-out.

panel retains the full range, including severe failures from L1 normalization ($-14.680/ -15.055$ pp) and $C = 0.1$ ($-12.000/ -12.563$ pp); the right panel magnifies the near-baseline region instead of letting those failures flatten all smaller effects.

Several configurations improve both recorded splits, but the ordering is not identical. $C = 2$ changed F1 by $+1.159/ +1.416$ pp and $\text{min_df} = 3$ by $+0.891/ +0.537$ pp on dev/held-out. In contrast, removing vector normalization changed F1 by $-1.021/ +1.219$ pp, while balanced class weights changed it by $+1.327/ +0.138$ pp. These reversals, despite Pearson $r = 0.977758$ and Spearman $\rho = 0.809146$ across the historical records, show why dev ranking cannot be treated as a reliable held-out ranking. The plot is a discrepancy-diagnosis ledger created after the frozen baseline missed the reference; it must not be used to replace that frozen Ticket 1 baseline or to claim a held-out-blind model search.

The row-level probes show boundary sensitivity without supplying a causal explanation. Dev ID 5282 changed from a correct positive at 0.500826 under the baseline to a false negative at 0.498597 under `liblinear`. The movement is consistent with a solver-dependent boundary change, but one row does not establish which configuration produced the undisclosed reference.

C. Ticket 2: Text Normalization

Hypotheses and lever. Ticket 2 asked whether canonicalizing URLs, mentions, hashtags, punctuation, case, or emoji would remove nuisance variation without erasing useful evidence. Each scored candidate enabled exactly one transformation; the raw control reproduced Ticket 1. The intended

decision required both dev evidence and a perturbation mechanism, rather than a score alone.

Dev evidence and decision. Table IX shows that URL placeholder produced the largest normalization F1, 0.740313 versus 0.738812 for raw text. Precision increased from 0.790941 to 0.804659 and accuracy from 0.789232 to 0.793171, while recall decreased from 0.693130 to 0.685496. Relative to Ticket 1 it fixed 22 FPs and 6 FNs, while introducing 11 FPs and 11 FNs. More importantly, on all 767 dev rows subjected to the paired URL substitution, the URL-normalized model changed zero predictions whereas the raw control changed 275. This alignment between the intended lever and measured invariance supported adoption.

Mention placeholder reduced F1 to 0.736756. Punctuation-to-space gained only 0.000602. Hashtag-marker removal, casefolding, and emoji placeholder caused no label or metric change. The emoji detector matched zero dataset rows, so this is absence of coverage, not evidence that emoji are useless.

Held-out evidence. After freezing, URL placeholder achieved precision 0.825137, recall 0.692661, F1 0.753117, and accuracy 0.804990 (Table V). It fixed 22 FPs and 8 FNs but introduced 4 FPs and 15 FNs (Table VII). Thus the held-out result preserved the dev mechanism: fewer false positives and higher precision, with a recall cost. The gain remains small, comes from one split, and does not measure the deployment frequency or value of link destinations.

D. Ticket 3: Shortcut and Shallow-Feature Audit

Hypotheses and lever. Ticket 3 isolated keyword, location, length, missingness, and surface-count blocks, then combined declared subsets with text. The hypothesis was that metadata

TABLE VIII
TICKET 1 HISTORICAL PROBE DELTAS ON DEV AND HELD-OUT

probe label	cause category	dev f1 target 1	heldout f1 target 1	dev delta percentage points	heldout delta percentage points
Frozen baseline control	frozen baseline control	0.738812	0.749186	0	0
Explicit unigram control	changed TF-IDF parameters	0.738812	0.749186	0	0
Add word bigrams	changed TF-IDF parameters	0.728745	0.726979	-1.006710	-2.220667
Add word trigrams	changed TF-IDF parameters	0.720387	0.717949	-1.842526	-3.123695
Disable sublinear TF	changed TF-IDF parameters	0.738812	0.749186	0	0
max_features=20,000	changed TF-IDF parameters	0.738812	0.749186	0	0
max_features=None	changed TF-IDF parameters	0.738812	0.749186	0	0
min_df=2	changed TF-IDF parameters	0.743525	0.755187	0.471344	0.600105
min_df=3	changed TF-IDF parameters	0.747725	0.754561	0.891335	0.537486
min_df=5	changed TF-IDF parameters	0.745033	0.751236	0.622107	0.204992
max_df=0.9	changed TF-IDF parameters	0.738812	0.749186	0	0
Disable IDF	changed TF-IDF parameters	0.723439	0.729184	-1.537328	-2.000216
Disable IDF smoothing	changed TF-IDF parameters	0.739414	0.746732	0.060164	-0.245364
L1 normalization	changed TF-IDF parameters	0.592016	0.598639	-14.679646	-15.054621
No vector normalization	changed TF-IDF parameters	0.728606	0.761373	-1.020569	1.218704
Unicode accent stripping	changed preprocessing	0.737785	0.749186	-0.102703	0
Single-character tokens	changed preprocessing	0.740501	0.743548	0.168917	-0.563728
Preserve case	changed preprocessing	0.731148	0.723265	-0.766450	-2.592036
C=0.1	regularization	0.618817	0.623552	-11.999536	-12.563354
C=0.25	regularization	0.704663	0.697998	-3.414883	-5.118741
C=0.5	regularization	0.726672	0.722269	-1.213987	-2.691711
C=2	regularization	0.750403	0.763347	1.159053	1.416095
C=4	regularization	0.748800	0.762658	0.998796	1.347256
C=10	regularization	0.745628	0.762276	0.681594	1.309025
class_weight=balanced	class weighting	0.752085	0.750569	1.327287	0.138294
Solver=liblinear	Logistic Regression solver	0.737785	0.748576	-0.102703	-0.060959
Explicit solver=lbfgs	Logistic Regression solver	0.738812	0.749186	0	0
max_iter=1,000	maximum iterations or convergence	0.738812	0.749186	0	0
random_state=1	random seed	0.738812	0.749186	0	0
random_state=42	random seed	0.738812	0.749186	0	0
random_state=9,999	random seed	0.738812	0.749186	0	0
Explicit L2 representation	package-version behavior	0.738812	0.749186	0	0
Leaky train+dev TF-IDF (invalid)	accidental leakage	0.725237	0.736932	-1.357518	-1.225336

Each delta is probe F1 minus submitted-baseline F1 in percentage points. The historical held-out ledger is diagnostic after the reference mismatch and is not eligible evidence for replacing the frozen baseline.

TABLE IX
TICKET 2 NORMALIZATION AND INVARIANCE RESULTS ON DEV

variant	f1 target 1	fixed fp	fixed fn	new fp	new fn	robustness affected rows	robustness changed predictions	robustness invariance rate
raw_text_control	0.738812	0	0	0	0	—	—	—
normalize_urls_placeholder	0.740313	22	6	11	11	767	0	1
normalize_mentions_placeholder	0.736756	2	1	2	3	399	0	1
strip_hashtag_marker	0.738812	0	0	0	0	330	0	1
normalize_punctuation_to_space	0.739414	1	2	0	2	1445	0	1
unicode_casefold	0.738812	0	0	0	0	1523	0	1
normalize_emoji_placeholder	0.738812	0	0	0	0	0	0	1

All variants use the frozen baseline model family; robustness values refer to each transformation’s matched perturbation. Emoji affected zero dataset rows.

might contain real benchmark signal while also encoding collection artifacts unlikely to remain stable outside the benchmark. A predeclared trust rule allowed rejection of the highest dev score if targeted perturbations exposed strong shortcut dependence.

Dev evidence and decision. Keyword-only F1 was 0.659859, length-only 0.437613, location-only 0.224414, and selected-shallow-only 0.530290 (Table X). Text plus keyword scored 0.735084, below text-only. Text plus keyword and selected shallow features reached the largest visible value, 0.749396, fixing 41 FPs and 46 FNs while introducing 42 FPs and 35 FNs over 164 changed predictions.

That aggregate gain failed the robustness criterion. Masking keyword changed 702 predictions and reduced the rich candidate’s F1 to 0.642306. Neutralizing superficial text changed 82 predictions and reduced F1 to 0.733068. For comparison, the text-only model under the same superficial neutralization scored 0.721053. The results show association with keyword availability, not a causal decomposition of feature importance;

mask-to-missing is also a severe stress test. Nevertheless, the dependence was strong enough to reject the rich candidate and retain the Ticket 1 text-only model. Ticket 3 therefore reused Ticket 1 held-out predictions without refitting.

The stable-ID cases expose the shortcut mechanism. Keyword-only assigned the same 0.663488 score to dev ID 7, a positive wildfire report whose missing-keyword representation corrected a baseline FN, and ID 25, the negative text “Summer is lovely,” where the same representation created an FP. Rich-model ID 331, a figurative concert description containing “annihilated,” moved from a correct negative score 0.418088 to an FP at 0.521747. These examples do not estimate prevalence, but they explain why the higher score was not accepted automatically.

E. Ticket 4: Decision Rule and Model

Hypotheses and lever. Ticket 4 held the raw-text TF-IDF representation fixed while evaluating 61 thresholds from 0.20 through 0.80, seven unweighted Logistic Regression values

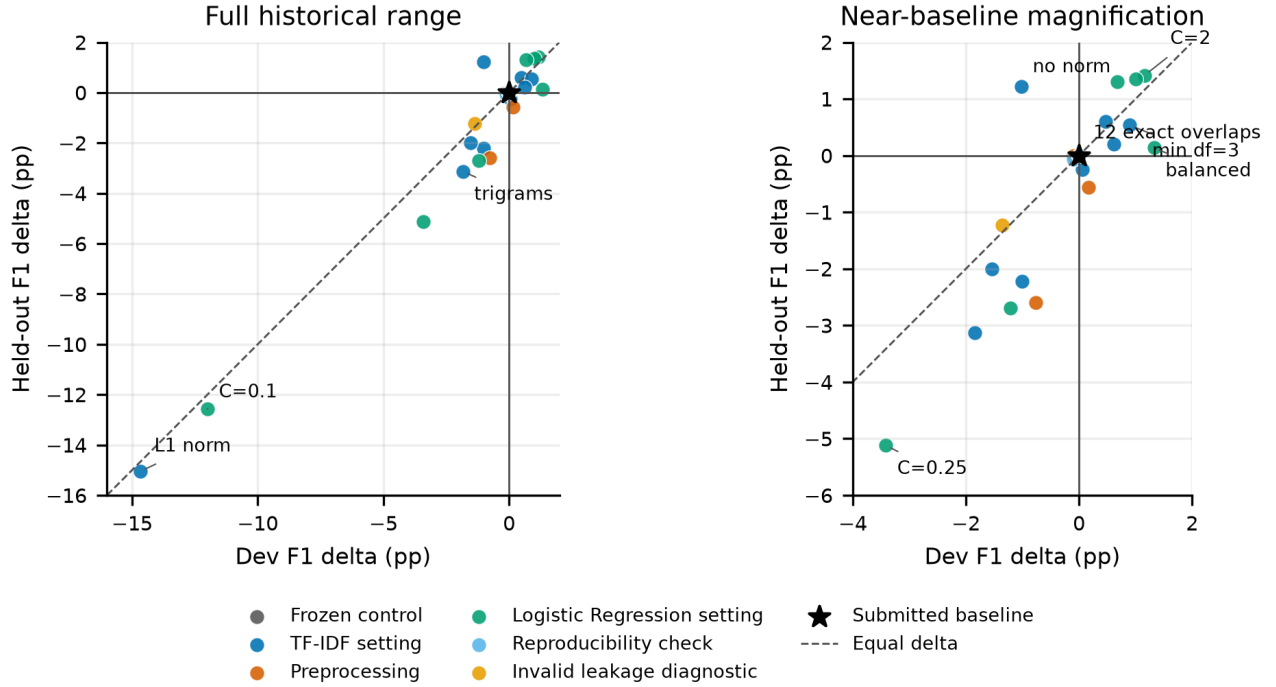


Fig. 1. Ticket 1 historical probe deltas. Each point is the target-1 F1 difference from the submitted baseline, in percentage points, with dev on the horizontal axis and held-out on the vertical axis. The left panel preserves extreme failures; the right panel magnifies near-baseline behavior. The 12 exact overlaps at (0, 0) are distinct baseline-equivalent configurations. These post-mismatch held-out records are descriptive diagnostics, not selection evidence.

TABLE X
TICKET 3 SHORTCUT AND SHALLOW-FEATURE AUDIT ON DEV

variant	f1 target 1	prediction changes	keyword mask changed predictions	keyword mask f1	neutralized changed predictions	neutralized f1
train_majority_floor	0	574	—	—	—	—
raw_text_tfidf_logistic_regression	0.738812	0	—	—	99	0.721053
keyword_only	0.659859	378	905	0.601469	—	—
length_only	0.437613	515	—	—	297	0.249110
keyword_plus_length	0.665060	338	929	0.601656	125	0.620567
location_only	0.224414	548	—	—	—	—
keyword_plus_location	0.658635	380	902	0.603263	—	—
selected_shallow_only	0.530290	442	1003	0.605042	242	0.520158
text_plus_keyword	0.735084	148	468	0.680352	77	0.714044
text_plus_selected_shallow_features	0.749396	164	702	0.642306	82	0.733068

Candidate metrics and matched perturbation outcomes are kept on dev. Blank robustness cells mean the perturbation was not applicable to that variant.

$C \in \{0.25, 0.5, 1, 2, 4, 5, 10\}$, balanced Logistic Regression at $C \in \{1, 5, 10\}$, and one LinearSVC comparator at $C = 1$. The threshold candidate uses the unweighted $C = 1$ scores; all other candidates use their native default rule. Lower thresholds and balanced weighting were expected to raise recall at a precision cost, while regularization and LinearSVC could change sparse-feature separation.

Dev evidence and decision. Figure 2 shows the expected precision–recall exchange across the baseline threshold sweep. The best threshold-only candidate was 0.47 with F1 0.749407. Figure 3 and Table XI show that regularization was not monotonic: unweighted $C = 2$ reached F1 0.750403 and the highest accuracy, 0.796454, while unweighted $C = 5$ and $C = 10$ reached 0.749403 and 0.745628. The balanced $C = 10$ extension reached F1 0.750958 with precision 0.753846 and recall 0.748092, but did not exceed balanced $C = 1$. LinearSVC achieved F1 0.740859.

The pre-execution plan records that the four $C = 5/10$ rows were historically motivated by a parallel V1 held-out observation. They are therefore reported as exploratory sensitivity evidence, not as an independent confirmation of the selection rule. Crucially, their inclusion did not determine the final choice: balanced $C = 1$ remains the dev-F1 maximum both in the original eight-candidate core and across the full recorded 12-candidate grid. It achieved F1 0.752085, precision 0.746988, and recall 0.757252, fixed 42 baseline FNs, and introduced 48 FPs. It was frozen at $C = 1$, balanced weights, and inclusive threshold 0.50.

Held-out evidence. The balanced model achieved precision 0.744361, recall 0.756881, F1 0.750569, and accuracy 0.783979. Relative to Ticket 1, F1 increased only 0.001383, precision fell 0.057033, recall rose 0.053517, and accuracy fell 0.013789. It fixed 35 FNs and introduced 56 FPs, with no fixed FPs or new FNs. This is a defensible F1-oriented, recall-heavy

TABLE XI
TICKET 4 MODEL AND DECISION-RULE COMPARISON ON DEV

variant	classifier family	class weight	decision threshold	precision target 1	recall target 1	f1 target 1	accuracy	fixed fn	new fp
lr_c1_unweighted_default	logistic_regression	—	0.500000	0.790941	0.693130	0.738812	0.789232	0	0
lr_c025_unweighted_default	logistic_regression	—	0.500000	0.811133	0.622901	0.704663	0.775443	5	6
lr_c05_unweighted_default	logistic_regression	—	0.500000	0.791367	0.671756	0.726672	0.782666	3	4
lr_c2_unweighted_default	logistic_regression	—	0.500000	0.793867	0.711450	0.750403	0.796454	16	9
lr_c4_unweighted_default	logistic_regression	—	0.500000	0.786555	0.714504	0.748800	0.793828	22	22
lr_c5_unweighted_default	logistic_regression	—	0.500000	0.782392	0.719084	0.749403	0.793171	25	25
lr_c10_unweighted_default	logistic_regression	—	0.500000	0.777778	0.716031	0.745628	0.789888	28	34
lr_c1_balanced_default	logistic_regression	balanced	0.500000	0.746988	0.757252	0.752085	0.785292	42	48
lr_c5_balanced_default	logistic_regression	balanced	0.500000	0.751543	0.743511	0.747506	0.783979	37	47
lr_c10_balanced_default	logistic_regression	balanced	0.500000	0.753846	0.748092	0.750958	0.786605	41	51
linear_svc_c1_default	linear_svc	—	0	0.772803	0.711450	0.740859	0.785949	27	43
lr_c1_unweighted_tuned_threshold	logistic_regression	—	0.470000	0.777049	0.723664	0.749407	0.791858	20	16

All rows preserve the frozen text representation; only the listed classifier/regularization/class-weight/threshold lever changes.

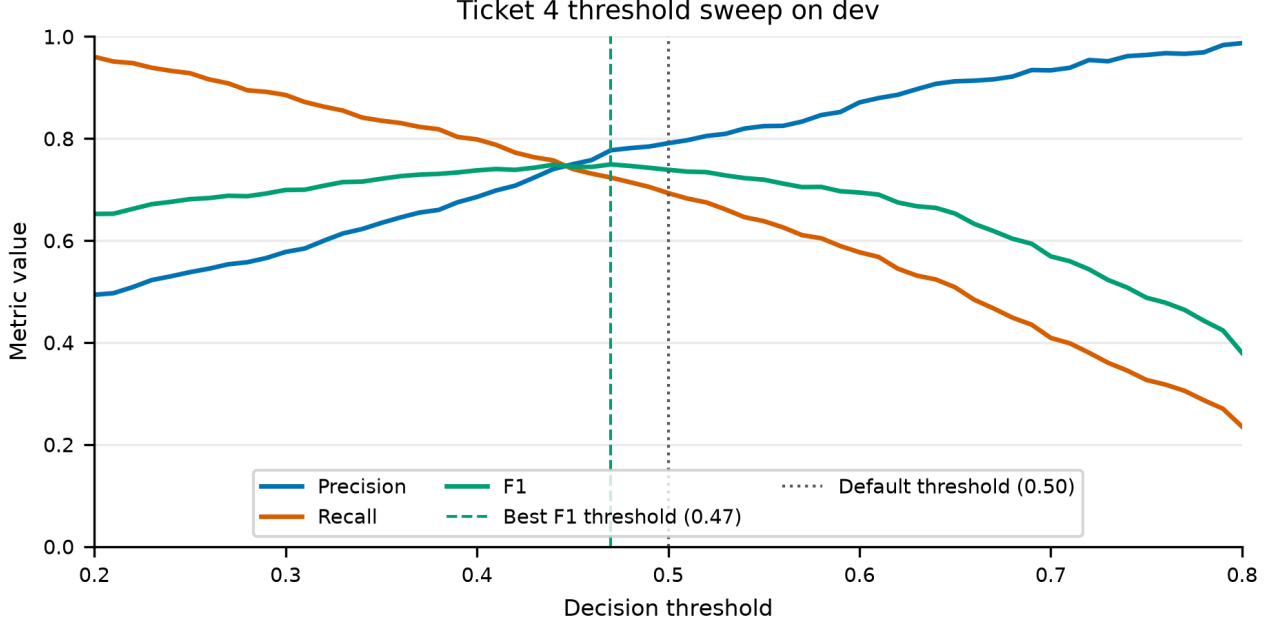


Fig. 2. Ticket 4 dev threshold sweep for the unweighted baseline scores. Precision increases and recall decreases as the threshold rises; F1 peaks at 0.47. The balanced model was evaluated separately and selected at the default threshold 0.50. The plotted source contains all 61 verified thresholds.

operating point under the declared rule, not an unequivocally superior classifier. No deployment cost function determines whether 35 recovered positives outweigh 56 new false alarms.

F. Ticket 5: Data Quality and Error Audit

Hypotheses and lever. Ticket 5 asked whether duplicate relationships, likely mislabels, ambiguity, hard negatives, or model weaknesses limited score interpretation, and whether a bounded correction intervention should alter the frozen model. Raw-exact conflicts were treated as evidence of annotation inconsistency. URL-canonical and bounded character-ngram near relationships were candidate generators requiring review. Model disagreement alone was never treated as proof of a wrong label. The only model lever was a single eight-row, train-only label proposal set tested in memory; dev and held-out labels remained untouched.

Dev evidence and decision. On train plus dev, the audit found 50 exact groups with 124 members, 13 conflicting groups, and 19 cross-split groups; 214 canonical groups with

677 members, 46 conflicts, and 98 cross-split groups; and 693 near pairs covering 538 unique IDs, with 104 conflicting and 265 cross-split pairs (Table XII). The eight-row correction candidate achieved precision 0.742129, recall 0.755725, F1 0.748865, accuracy 0.782009, and 696/172/160/495 confusion counts. It reduced F1 by 0.003220, failing the predeclared -0.002 noninferiority margin. It fixed 3 errors and introduced 8—7 FPs and 1 FN—so it also failed the error-balance gate. The decision was to retain Ticket 4 with an empty correction list.

Post-freeze audit evidence. After freeze, full-data discovery found 69 exact groups/179 members, 292 canonical groups/932 members, and 960 near pairs covering 715 unique IDs. Cross-split counts were 41, 194, and 546 respectively. These categories overlap and near pairs are candidates, so counts must not be added as unique leakage totals. The final 64-row audit contains 30 fix, 15 ambiguous, 6 keep_but_flag, and 13 reject_false_positive

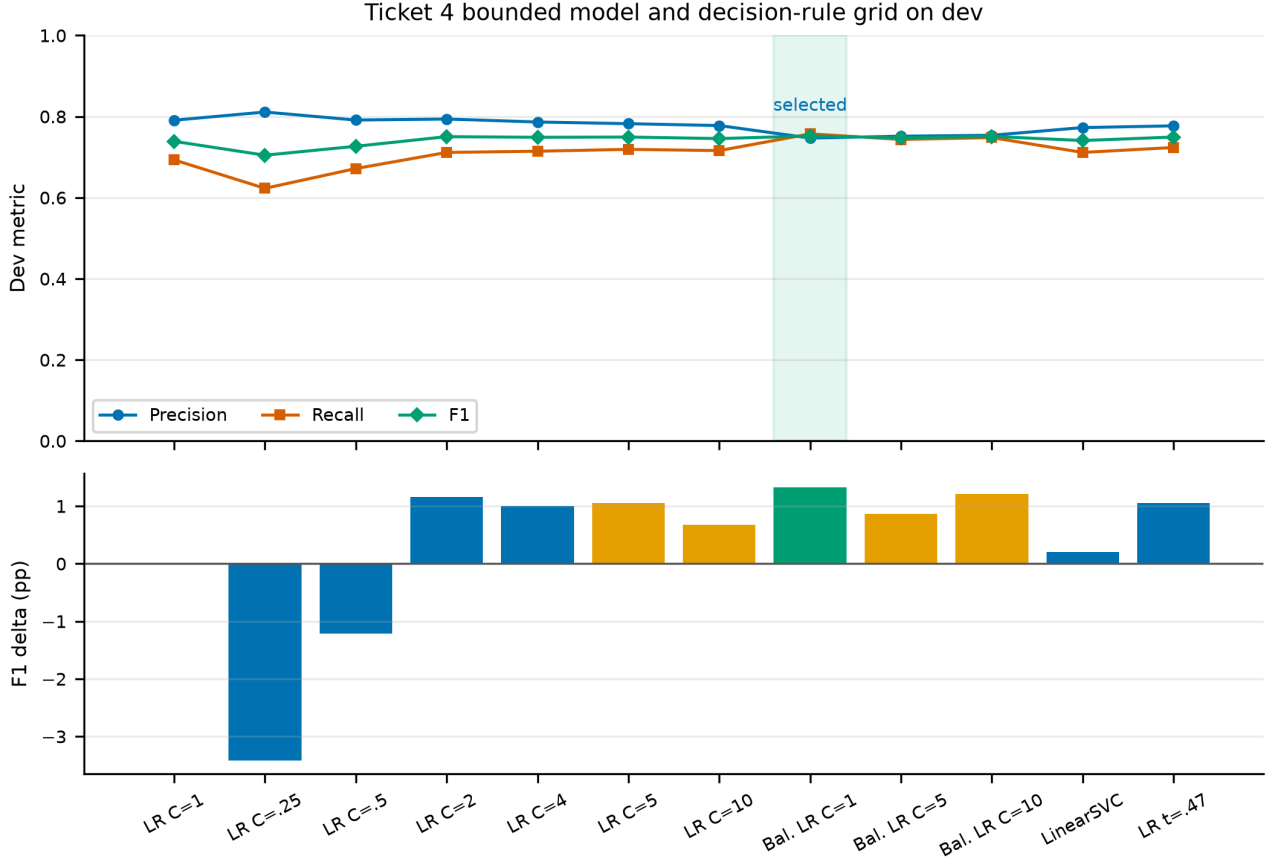


Fig. 3. Ticket 4 dev model grid. The upper panel preserves the full $[0, 1]$ metric scale; the lower panel exposes F1 changes in percentage points relative to unweighted $C = 1$. Green marks the frozen selection and orange marks the $C = 5/10$ extensions. The extensions are reported as sensitivity evidence because the plan records a historical V1 held-out observation as their motivation.

TABLE XII
TICKET 5 DATA-QUALITY AUDIT SUMMARY

scope	relationship type	groups or pairs	member rows	conflicting groups or pairs	cross split groups or pairs
train_plus_dev	exact	50	124	13	19
train_plus_dev	canonical	214	677	46	98
train_plus_dev	near	693	538	104	265
full_with_heldout	exact	69	179	18	41
full_with_heldout	canonical	292	932	64	194
full_with_heldout	near	960	715	186	546
audit_disposition	ambiguous	—	15	—	—
audit_disposition	fix	—	30	—	—
audit_disposition	keep_but_flag	—	6	—	—
audit_disposition	reject_false_positive	—	13	—	—

Duplicate rows report groups/pairs and member tweets; disposition rows report audited tweet counts. Source and held-out labels remain unchanged.

dispositions. “Fix” means a recommendation only. The audit manifest confirms that no source or held-out label was changed and no held-out row was removed.

therefore Ticket 5: the Ticket 4 balanced Logistic Regression recipe with no training-label corrections.

G. Frozen Final Choice and Cross-Ticket Interpretation

Figure 4 compares dev and held-out F1 without changing the decision rule. Tickets 1 and 3 share one prediction core; Tickets 4 and 5 share the final core. Ticket 2 has the largest observed held-out F1, but using that observation to replace Ticket 5 would be held-out tuning. The valid final identity is

Figure 5 makes the semantic cost clearer than F1 alone. URL placeholder mainly removes FPs but also creates FNs. Balanced weighting recovers FNs at the cost of more new FPs. A single net error count would conceal this directional difference.

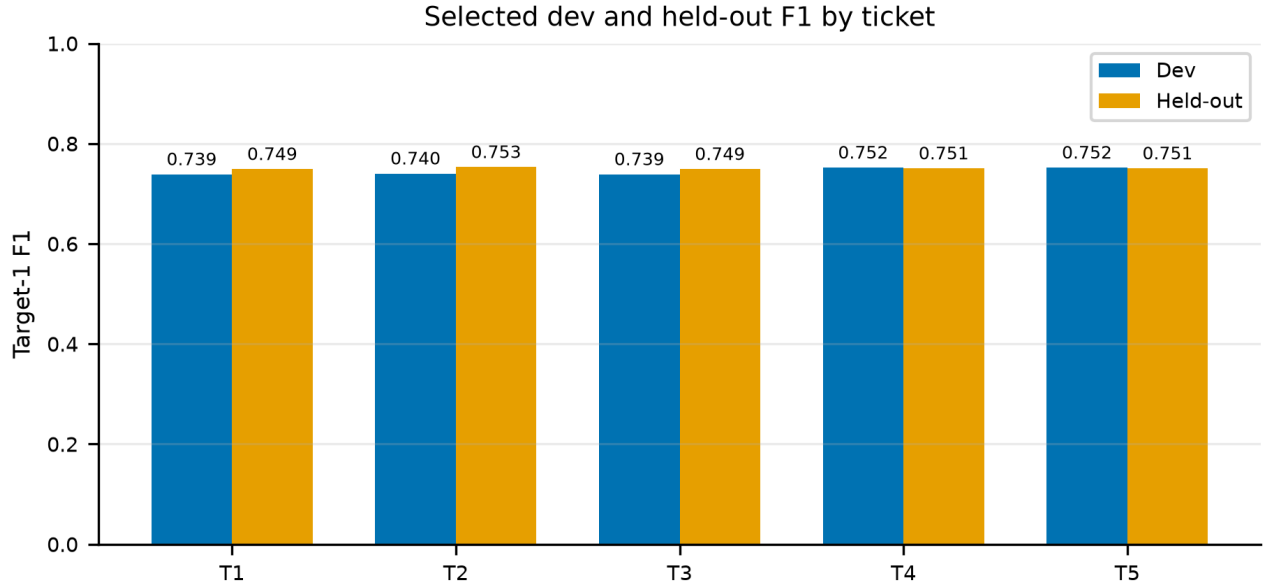


Fig. 4. Selected dev and post-freeze held-out target-1 F1 by ticket. The full $[0, 1]$ axis avoids exaggerating small differences. T1/T3 and T4/T5 are intentional shared prediction cores; held-out bars are descriptive rather than a new selection criterion.

IV. CASE ANALYSIS

A. Prediction Movements that Match Intended Levers

Table XIII deliberately contains favorable and unfavorable cases. For URL placeholder, held-out ID 773 is a target-0 advertising tweet with two links. Its baseline FP score 0.533045 falls to 0.460680, fixing the error. Conversely, ID 902 is a target-1 bioterrorism report whose score falls from 0.508666 to 0.499356, creating an FN. The pair is consistent with a transformation that removes link-string sensitivity but can move borderline positives below threshold.

For balanced weighting, held-out ID 556 is a target-1 arson-related example whose score rises from 0.464323 to 0.529717, fixing an FN. ID 59 is a target-0 link-heavy post whose score rises from 0.450541 to 0.504457, creating an FP. These cases instantiate the recall-precision exchange in Table VII. ID 767 is another verified new FP. Its machine-generated scores are 0.487623 and 0.544815; older narrative values conflict with the transition CSV and are not used.

The Ticket 5 correction probe is similarly mixed. Dev ID 9470 moves from an FP at 0.633044 to a correct negative at 0.465278, but ID 4014 moves from a correct negative at 0.496509 to an FP at 0.502209. The aggregate gate correctly prevented a small set of plausible semantic edits from being presented as an unqualified model improvement.

B. Hard Negatives, Potential Annotation Issues, and Ambiguity

Error direction does not adjudicate a label. Held-out ID 5760 explicitly says there are “no forest fires.” The final model predicts positive at 0.768287, but the audit rejects the suspicion that the negative label is wrong; this is a valid hard negative and a model error involving negation. ID 10795, a

target-1 report about a wrecked home, is predicted negative at 0.135000; the audit likewise treats it as model weakness rather than a mislabel.

By contrast, held-out ID 2619 is labeled target 1 but says “My iPod crashed” with fan hashtags. The audit recommends a potential 1-to-0 fix, but no external adjudicator confirmed it and the source label remains unchanged. IDs 6220 and 6223 contain identical text about the historical D. B. Cooper image but have opposite held-out labels. Since the evidence is identical, both are marked ambiguous and unsuitable for automatic correction. These cases show why the four-way audit vocabulary separates recommendation, ambiguity, flagging, and rejection.

Figure 6 summarizes the curated audit, not the prevalence of defects in all 7,613 rows. Its 64 records were selected for review and the near-duplicate search was bounded. The semantic judgments are human/AI-supported assessments without independent expert or multi-annotator ground truth.

C. Why Aggregate Improvement Is Insufficient

The case and robustness evidence support a consistent conclusion. Ticket 2’s small gain accompanies both fixed and new errors. Ticket 3’s largest visible dev score depends strongly on keyword availability. Ticket 4’s F1 choice recovers positives while adding more false alarms and lowering accuracy. Ticket 5 shows that apparently plausible corrections can reduce agreement with the benchmark. Thus a higher F1 can coexist with shortcut fragility, undesirable operating costs, or contradictory labels. Trustworthiness in this report means convergence among controlled dev selection, expected directional movements, targeted robustness, stable-ID exam-

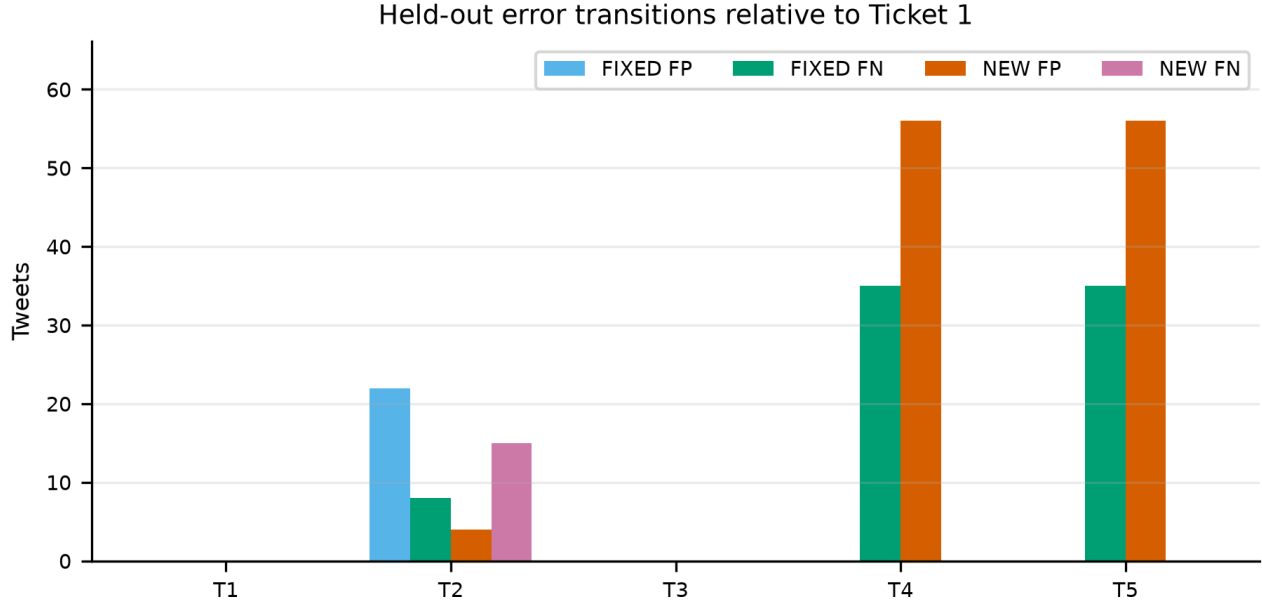


Fig. 5. Held-out fixed and newly introduced error counts relative to Ticket 1. Counts begin at zero and remain separated by error type.

TABLE XIII
REPRESENTATIVE SUCCESSES, FAILURES, AMBIGUITIES, AND AUDIT CASES

ticket	split	id	case type	true label	baseline pred	baseline score	candidate pred	candidate score	transition or disposition
1	dev	5282	solver boundary failure	1	1	0.500826	0	0.498597	new_error
2	heldout	773	normalization success	0	1	0.533045	0	0.460680	fixed_error
2	heldout	902	normalization failure	1	1	0.508666	0	0.499356	new_error
3	dev	7	shortcut success	1	0	0.412138	1	0.663488	fixed_error
3	dev	25	shortcut failure	0	0	0.265330	1	0.663488	new_error
3	dev	331	shallow-feature failure	0	0	0.418088	1	0.521747	new_error
4	heldout	556	recall success	1	0	0.464323	1	0.529717	fixed_error
4	heldout	59	precision failure	0	0	0.450541	1	0.504457	new_error
4	heldout	767	verified score conflict	0	0	0.487623	1	0.544815	new_error
5	dev	9470	correction-probe success	0	1	0.633044	0	0.465278	fixed_error
5	dev	4014	correction-probe failure	0	0	0.496509	1	0.502209	new_error
5	heldout	5760	validated model error	0	—	—	1	0.768287	reject_false_positive
5	heldout	2619	potential annotation issue	1	—	—	0	0.224300	fix
5	heldout	6220	ambiguity	0	—	—	1	0.645703	ambiguous
5	heldout	6223	ambiguity	1	—	—	1	0.645703	ambiguous
5	heldout	10795	validated model error	1	—	—	0	0.135000	reject_false_positive

The CSV preserves full text, scores, and rationales; the compact report table preserves the verified numeric case ledger. Stable IDs support exact lookup.

ples, and freeze provenance. It is not a formal statistical proof or deployment guarantee.

V. DIFFICULTIES AND SOLUTIONS

Table XIV records six difficulties supported by machine artifacts. Three were especially consequential.

First, the literal baseline missed the reference but the reference exposed too little evidence to isolate a cause. The response was to verify ID alignment, train-only fitting, convergence, effective settings, seeds, versions, and 32 one-lever probes, then preserve the mismatch as unresolved rather than invent a causal explanation. The later Ticket 1 artifact reconstruction was qualified as one exact frozen-recipe recovery replay, not another selection round. The complete prior deletion history cannot be independently reconstructed because usable Git history is absent.

Second, the largest Ticket 3 dev gain appeared attractive until targeted masking exposed dependency on keyword availability. Adding robustness probes converted a score comparison into an evidence-based rejection. This solution remains bounded: the probes cover predefined feature families and severe perturbations, not every possible shortcut.

Third, Ticket 4 required selecting an operating point without held-out tuning. The solution was a predeclared dev criterion, a complete 61-threshold curve, bounded classifier candidates, confusion matrices, and fixed/new error analysis before freezing. The resulting choice is described as recall-oriented because the artifacts show both its benefit and cost.

Data quality created a fourth difficulty: duplicate conflicts, ambiguity, and model weakness cannot be separated by prediction error alone. Exact/canonical/near discovery was therefore separated from four allowed dispositions, and label proposals

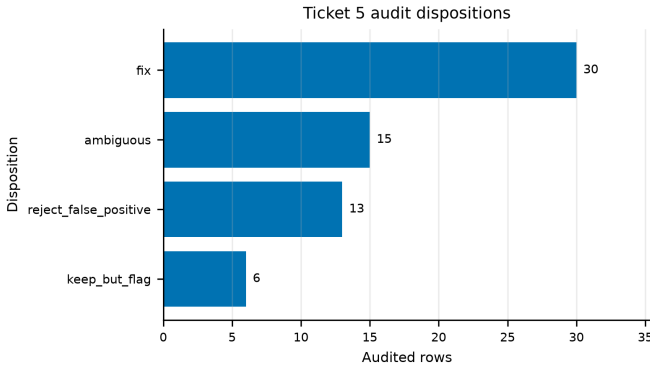


Fig. 6. Disposition counts in the 64-row data-quality audit. “Fix” denotes a recommendation, not an applied label mutation.

were tested only in memory under a dev gate. Finally, replay produced maximum probability drift of 1.11×10^{-16} despite identical labels and metrics. Verification used a 10^{-12} score tolerance, exact stable-ID/label/order checks, and a 10^{-15} metric tolerance rather than falsely requiring cross-library byte identity.

VI. AI USAGE DECLARATION

OpenAI Codex was used as the repository-operating AI assistant [11]. No retained evidence establishes the use of an independent labeling agent or external expert service. Codex assisted with repository inspection, requirement decomposition, implementation planning, Python and test drafting, controlled experiment orchestration after user authorization, artifact review, reproducibility auditing, and report preparation. The retained interaction record is a curated reconstruction supported by files and commands; it is not a verbatim export of all messages or hidden model reasoning.

AI-generated material was not accepted automatically. Code and experiment suggestions were accepted only when executable tests and artifact checks passed. Interpretations were changed or rejected when row-level evidence disagreed: for example, the stale narrative scores for ID 767 were rejected in favor of the machine transition CSV, unsupported explanations for the Ticket 1 reference gap were omitted, and semantic Ticket 5 “fix” judgments remained recommendations rather than source edits. Higher-scoring empirical candidates were also rejected when the predeclared evidence rules failed, as in Ticket 3’s shortcut-sensitive rich model and Ticket 5’s correction probe.

Verification included schema and stable-ID tests; train/dev/hold-out separation checks; deterministic hashes; prediction-level recomputation of precision, recall, F1, accuracy, confusion counts, and transitions; freeze chronology; source-label integrity; and five clean-process replays. The retained validation record reports 56 passing tests, no broken package requirements, a non-refitting submission-validator PASS, five reproduced ticket cores, and byte-exact regeneration of the three active result tables. The report-asset generator added 206 assertions over source hashes, archived

prediction hashes, schemas, row counts, stable IDs, labels, configurations, metrics, cases, and source paths.

Human responsibility remains explicit. The user authorized major stages and held-out/audit actions. AI suggestions did not provide independent ground truth. The submitter must verify the final author metadata, citations, wording, and course-policy compliance, and must review semantic audit judgments before submission. Table XV summarizes this division of labor.

VII. DISCUSSION AND LIMITATIONS

A. What Evidence Is Trustworthy

The strongest evidence in this project combines several independent constraints. Ticket 2 was adopted because a small dev gain aligned with a directly measured URL-invariance mechanism and a precision-oriented transition pattern that persisted after freeze. Ticket 3 was rejected because its higher score conflicted with robustness evidence. Ticket 4 was adopted cautiously because it won the declared dev criterion and its predicted recall mechanism persisted, although its held-out F1 advantage was small. Ticket 5 rejected corrections because both the noninferiority and error-balance gates failed. These different decisions demonstrate that “highest score” and “most trustworthy evidence” are not synonyms.

Internal reproducibility is strong within the locked environment. Five isolated processes reproduced all selected labels; maximum score drift was 1.11×10^{-16} , below the 10^{-12} tolerance. The summary, threshold sweep, and data-quality audit regenerated byte-for-byte. Final predictions contain 1,523 unique held-out IDs in the fixed order. This supports the provenance of the reported observations, but it does not validate the benchmark labels or prove cross-platform bit identity.

B. Remaining Uncertainty and Risks

The Ticket 1 discrepancy remains causally unresolved because the supplied reference is a scalar without prediction-level or configuration evidence. Small differences among candidates are based on one dev split; the project contains no bootstrap, repeated split, confidence interval, or formal significance analysis. Text duplicates and approximate relationships violate an independence intuition even though stable IDs are disjoint. No duplicate-group-aware evaluation estimates how performance changes when related text is kept within one partition.

Ticket 3 perturbations are severe and benchmark-specific. Keyword coefficients and score movements are associations rather than causal importance measures. Ticket 4 explores a finite grid and one alternative classifier; it has no calibrated deployment cost for false alarms versus misses. Ticket 5’s canonicalization can hide URL-destination information, and its near-duplicate search uses at most seven non-self neighbors at one threshold and representation. The 64 semantic audit records are curated rather than exhaustive, and no expert multi-annotator adjudication exists. Links were not fetched, so destination context was not verified.

The hidden-test risk follows directly from these limits. Superficial URL, keyword, location, missingness, and length

TABLE XIV
VERIFIED PROJECT DIFFICULTIES, RESPONSES, AND RESIDUAL LIMITS

difficulty	verified evidence	response	residual limit
Assignment reference and frozen baseline did not match	Held-out absolute gap=0.00823649010418; tolerance=0.001.	F1 Ran isolated dev probes and structural checks, then retained the reproducible frozen implementation and documented the unresolved gap.	The assignment’s hidden reference pipeline remains unknown.
Score-level numerical drift under clean replay	Final replay audit records stable labels/order and bounded score drift.	Compared hashes, stable IDs, labels, metrics, and score differences under explicit tolerances.	Exact floating-point identity is not claimed.
Apparent dev gain depended on shortcut-prone features	Matched keyword masking and superficial-text neutralization materially changed predictions and F1.	Rejected the richer candidate and retained the text-only frozen baseline.	The audit covers predefined shallow features and perturbations, not every possible shortcut.
Recall gains created precision costs	Balanced Logistic Regression fixes false negatives while introducing false positives on dev and held-out.	Evaluated bounded candidates, a full threshold sweep, confusion matrices, and transition counts before freezing the default-threshold balanced model.	The operating point reflects target-1 F1, not a deployment cost function.
Duplicate conflicts and ambiguous annotations	Duplicate summaries and 64 stable-ID audit records include conflicts and four dispositions.	Kept source/held-out labels unchanged, separated fix/ambiguous/keep/reject dispositions, and gated a train-only correction probe on dev.	Audit dispositions are evidence assessments, not ground-truth adjudications.
Emoji normalization lacked dataset coverage	The matched robustness artifact reports zero affected dev rows.	Retained the zero-coverage result and did not infer a dataset-level benefit.	The emoji hypothesis remains unresolved on this split.
Only difficulties supported by generated artifacts are included; narrative-only events are excluded.			

TABLE XV
AI USAGE AND VERIFICATION WORKFLOW

workflow stage	ai assistance	verification	human accountability
Evidence and contract audit	Repository inspection, requirement decomposition, and artifact inventory	Claims cross-checked against handout, clarifications, generated CSV/JSON, predictions, and frozen manifests	Final interpretation and submission remain the student’s responsibility
Implementation and testing	Pipeline/test drafting and diagnostic support	Automated tests, deterministic split checks, artifact schemas, hashes, and row-level assertions	No AI statement is accepted when it conflicts with machine-generated artifacts
Experiment orchestration	Running authorized dev experiments and producing frozen artifacts	Selections use dev evidence; held-out completion files and frozen manifests enforce usage policy	Held-out results did not reopen ticket selection
Reproducibility audit	Clean-process replay and comparison of archived outputs	Five tickets reproduced with stable IDs, labels, metrics, transitions, and bounded score tolerances	Exact score identity is not overstated
Report asset generation	Programmatic table, figure, and case-material generation	This script recomputes metrics, verifies labels/IDs/configuration, and records an asset manifest	Generated assets must be reviewed before inclusion
This is a workflow declaration, not a substitute for source citations or human review.			

patterns can be perturbed without changing intended labels. The accepted URL transformation has a measured invariance benefit, whereas the rejected rich model shows substantial keyword sensitivity. Nevertheless, neither experiment proves robustness to all hidden variants, temporal drift, new disasters, changing platform conventions, or different label policies.

C. Final Model Interpretation and External Validity

The final model is the last valid dev-based frozen choice: raw word-unigram TF-IDF plus balanced Logistic Regression at $C = 1$ and threshold 0.50, with no label corrections. It is recall-oriented. Relative to Ticket 1 on held-out, recall increases from 0.703364 to 0.756881 while precision decreases from 0.801394 to 0.744361 and accuracy from 0.797768 to 0.783979. The F1 gain is only 0.001383. These observations justify the procedural choice under the declared F1 criterion, not a universal operational optimum.

A deployment decision would require explicit costs, external or temporal validation, duplicate-aware splitting, threshold calibration under the deployment base rate, and a clearer annotation policy. Ticket 2’s higher observed held-out F1 is a useful result, but choosing it post hoc would compromise the study’s held-out discipline. The report therefore preserves the final frozen identity rather than rewriting the conclusion around the most favorable visible score.

VIII. CONCLUSION

This forensic study found that the local raw-text TF-IDF plus Logistic Regression baseline is reproducible and leakage-controlled but does not match the undisclosed reference within tolerance. URL placeholderism is a defensible isolated surface lever because a small precision-oriented gain aligns with exact paired perturbation invariance. Shallow-feature augmentation is not adopted despite a higher dev F1 because the gain is strongly keyword-sensitive. Balanced Logistic Regression changes the operating point toward recall and wins the declared dev F1 criterion, but its held-out advantage is small, precision and accuracy fall, and new FPs exceed recovered FNs. Duplicate conflicts and ambiguous labels materially limit score interpretation, yet the tested eight-label correction set worsens dev evidence and is rejected.

The final frozen decision is therefore balanced Logistic Regression at the default threshold with no label corrections. The central lesson is methodological: aggregate scores become credible only when connected to controlled levers, fixed and newly introduced errors, robustness probes, stable-ID cases, frozen chronology, and reproducible artifacts. Even that evidence remains bounded by one benchmark, one fixed split, unresolved reference details, duplicate dependence, and unadjudicated labels.

REFERENCES

- [1] Kaggle, “Natural language processing with disaster tweets,” Kaggle competition dataset, data source documented in the repository README. [Online]. Available: <https://www.kaggle.com/c/nlp-getting-started>
- [2] UC Berkeley RISELab, “Kaggle nlp disasters public data mirror,” GitHub repository, public mirror specified by the project starter. [Online]. Available: <https://github.com/ucbrise/kaggle-nlp-disasters/tree/master/data>
- [3] “Final project—topic a: Text classification pipeline forensics,” Course project handout, repository file: `topic-a-handout.md`.
- [4] “Instructor clarifications: Floor model and prediction-transition comparator,” Course clarification document, repository file: `teacher_clarifications.md`.
- [5] “Topic a starter: Text classification pipeline forensics,” Course starter documentation, repository file: `starter/README.md`.
- [6] “Real starter data,” Course dataset and fixed-split documentation, repository file: `starter/data/README_DATA.md`.
- [7] scikit-learn developers, “scikit-learn, version 1.9.0,” Software package, version recorded in `requirements-lock.txt` and experiment `software_versions.json` artifacts. [Online]. Available: https://scikit-learn.org/stable/whats_new/v1.9.html
- [8] pandas development team, “pandas, version 3.0.3,” Software package, version recorded in `requirements-lock.txt` and experiment `software_versions.json` artifacts. [Online]. Available: <https://pandas.pydata.org/docs/whatsnew/v3.0.3.html>
- [9] NumPy developers, “Numpy, version 2.5.1,” Software package, version recorded in `requirements-lock.txt` and experiment `software_versions.json` artifacts. [Online]. Available: <https://numpy.org/doc/2.5/>
- [10] Matplotlib development team, “Matplotlib, version 3.11.1,” Software package, version recorded in `requirements-lock.txt` and experiment `software_versions.json` artifacts. [Online]. Available: <https://matplotlib.org/stable/>
- [11] OpenAI, “Codex,” AI coding assistant, use and verification are documented in `logs/chat.md`; the retained log is curated rather than verbatim. [Online]. Available: <https://learn.chatgpt.com/>

APPENDIX A

EXTENDED CONFIGURATIONS AND NEGATIVE RESULTS

Ticket 1’s complete probe artifacts show that lowercasing off, bigrams, `liblinear`, $C = 0.5$, and an intentionally leaky train-plus-dev vocabulary all reduced dev F1. Raising `max_iter`, changing the seed, and spelling L2 explicitly produced no label change. These negative and null results constrain explanations but do not reveal the undisclosed reference implementation.

Ticket 2’s full normalization table is Table IX. Mention placeholdering was negative; punctuation-to-space was only marginally positive; hashtag stripping, casefolding, and emoji placeholdering were label-equivalent to raw text. Ticket 3’s full ten-row comparison is Table X. The isolated keyword, location, length, and shallow families were all weaker than text-only, while the rich model’s visible gain failed robustness. Ticket 4’s complete 12-row model table is Table XI; $C = 0.25$ and $C = 0.5$ hurt, the $C = 5/10$ extensions did not surpass balanced $C = 1$, LinearSVC did not win, and both threshold 0.47 and unweighted $C = 2$ lost to balanced weighting under the declared dev criterion. The extended rows are exploratory because of their documented V1 motivation. These are bounded comparisons on one split, not exhaustive searches.

APPENDIX B

DATA-QUALITY AND CORRECTION DETAIL

Raw-exact relationships require identical stored text. Canonical relationships use deterministic text canonicalization, including URL handling, to identify closely related reposts. Near relationships are bounded character-ngram candidates rather than proof of equivalence. Categories overlap. Audit records use four dispositions: `fix` for a supported recommendation, `ambiguous` when evidence cannot resolve a label, `keep_but_flag` for a retained row with a documented concern, and `reject_false_positive` when a suspicion of data error is rejected.

The correction probe contained eight stable training IDs: 4076, 6566, 8698, 8739, 1723, 1760, 6097, and 9472. Original and proposed labels were preserved side by side, and only an in-memory training copy changed. The candidate failed its dev gates, so none of the proposals was applied to source data. Rejection of the candidate does not prove that all original labels are semantically correct; it shows only that the bundled intervention was not justified for the frozen classifier under the declared dev criteria.

APPENDIX C

STABLE-ID CASE AND EVIDENCE PROVENANCE

Table XIII is the report’s extended case ledger. Its CSV source preserves full text, probability values, and selection rationales; the compact report table preserves IDs, labels, predictions, scores, and transition or disposition. Cases were selected from actual transition or audit files and include successes and failures. They are illustrative, not prevalence estimates. Ticket 4 ID 767 uses only the machine-generated scores.

All report tables and figures were generated by `scripts/generate_report_assets.py`. The exact source list for each asset is `report_assets/asset_manifest.csv`, and the 206 assertion outcomes are in `report_assets/validation_report.json`. Central frozen recipes, hashes, archived prediction paths, expected metrics, and transition counts are in `configs/frozen_decisions.json`. Run commands remain in per-experiment `run_command.txt` files rather than being pasted into this report.

The clean audit recorded five distinct replay processes, identical labels and stable-ID order, maximum score drift 1.11×10^{-16} , and no reopened selection. It regenerated `results/summary.csv`, `results/threshold_sweep.csv`, and `results/data_quality_audit.csv` byte-for-byte. The final prediction file is `predictions/final-heldout-predictions.csv`. This establishes internal reproducibility under the locked environment; it does not establish bit-identical scores across arbitrary platforms.

APPENDIX D
REPORT FINALIZATION BOUNDARIES

The author names were supplied directly. Unsupplied affiliation, location, and email metadata are omitted rather than

inferred. Bibliographic URLs and software versions were checked against official pages where available; course-internal sources use repository filenames because no public metadata was supplied.